



Introduction to Spark

Installing Spark

Abstract

The goal of this guide is to install stand-alone Spark and the sample data files required for class. Spark and the data files are needed to complete the assignments for this class. This lab will require roughly 30 to 45 minutes to complete.

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Prerequisites

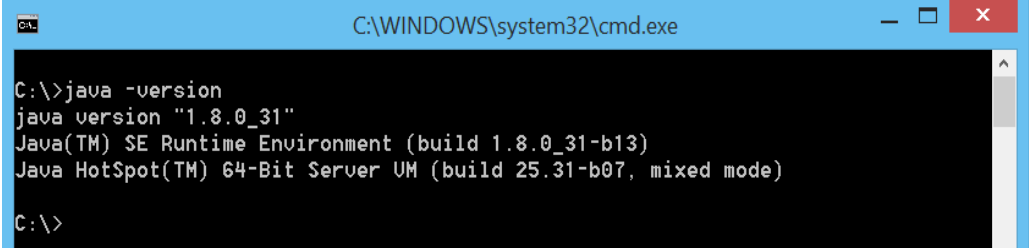
- ❑ A computer running Windows 7 or newer, 4 GB RAM, 10 GB free disk space is required. Windows 64-bit version recommended.

- ❑ Java Runtime Environment version 7 or higher, Version 8 is recommended.

<http://java.com/download>

- To check your Java version type the following command at the command prompt: **java -version**

The output should look like this:



```
C:\WINDOWS\system32\cmd.exe

C:\>java -version
java version "1.8.0_31"
Java(TM) SE Runtime Environment (build 1.8.0_31-b13)
Java HotSpot(TM) 64-Bit Server VM (build 25.31-b07, mixed mode)

C:\>
```

- the system response to the **java -version** command must be similar to the above

IMPORTANT: If you do not receive the above response, follow the instructions in the Appendix: Java installation troubleshooting.

- ❑ Archiving program which is able to handle Linux *.tar and *.gz archives is required.

- In this guide the free open-source tool 7-Zip application is used. The 7-Zip tool can be downloaded and installed from this link: <http://www.7-zip.org/> . Download and run the version for your architecture, whether 32-bit or 64-bit, and follow the on-screen instructions to install 7-Zip.



Download software packages and data

Standalone Spark requires Scala to be installed and configured before Spark can be run. Caution is needed to install the proper version of Scala.

IMPORTANT: Spark will only work properly with the version of Scala it was compiled against.

1. Visit <http://spark.apache.org/downloads.html> to download latest stable release of Spark:
 - a) Option “2. Choose a package type” select “Pre-built for Hadoop 2.6 and later”.
 - b) Option “3. Choose a download type” select “Select Apache Mirror”.
 - c) Select the link provided beside “4. Download Spark” and choose the recommended mirror.
2. Determine the required version of Scala. Under the “Link with Spark” section of the spark download page, you can find the Scala version of the default Spark release in the name of the Maven artifact. The Scala version is presented at the end of the artifactId line:

Link with Spark

Spark artifacts are hosted in Maven Central.

```
groupId: org.apache.spark  
artifactId: spark-core-2.10  
version: 1.3.0
```

NOTE: Default releases of Spark 1.3 and 1.4 need Scala version 2.10.

3. Go to <http://www.scala-lang.org/download/all.html> and download the latest release of the required version of Scala.
4. Download **winutils.exe** from the Hortonworks page: <http://public-repo-1.hortonworks.com/hdp-win-alpha/winutils.exe>.
5. Download training data from course Resources.

The screenshot shows a course interface with a list of resources on the left and a sidebar on the right. The list of resources includes:

- 3. Go to <http://www.scala-lang.org/download/all.html> and download the latest release of the required version of Scala.
- 4. Download **winutils.exe** from the Hortonworks page: <http://public-repo-1.hortonworks.com/hdp-win-alpha/winutils.exe>.
- 5. Download training data from course Resources.

The sidebar on the right contains a "View resources" button (circled in red) and a "Downloadable resources" section. In this section, there is a button labeled "Training Data" (circled in red) with a download icon.

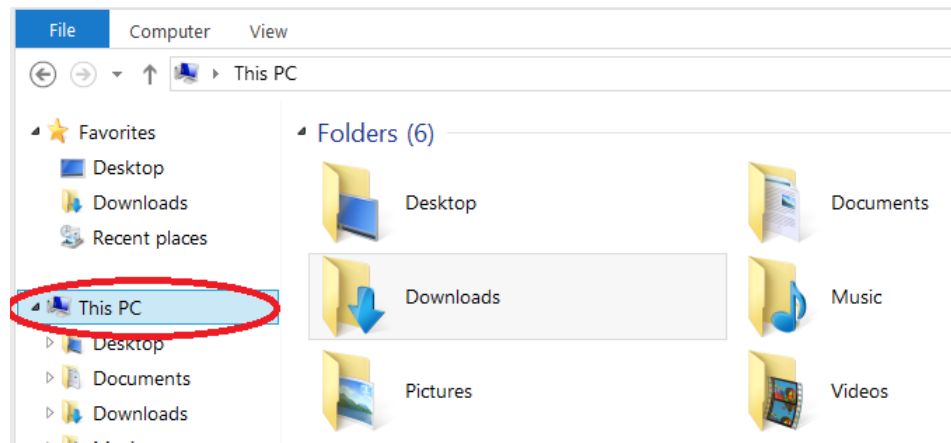


Install Scala

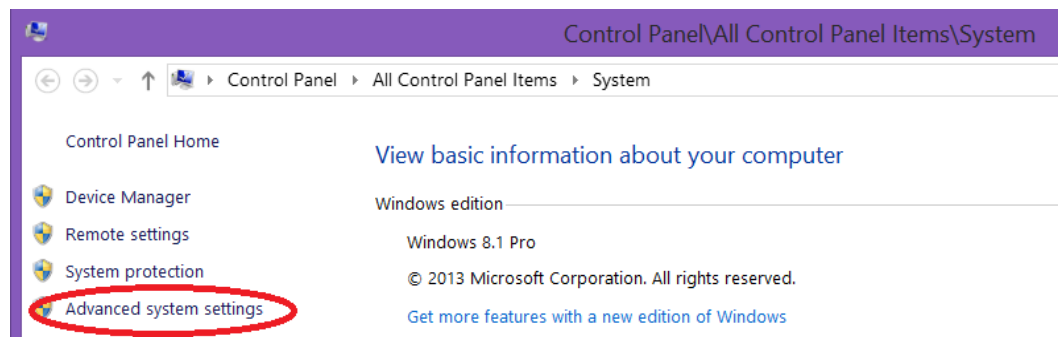
1. Uninstall any installed older/newer versions of Scala.
2. Run the Scala Installer downloaded in the previous section point 4. Follow the on-screen instructions. You do not need to change default installation options.
3. Add **SCALA_HOME** environment variable and add **%SCALA_HOME%\bin** into your **Path**

NOTE: The procedure below applies to Windows 8/8.1. For other systems please refer to your operating system documentation.

- a) In 'File Explorer', right-click 'This PC' icon and select 'Properties':

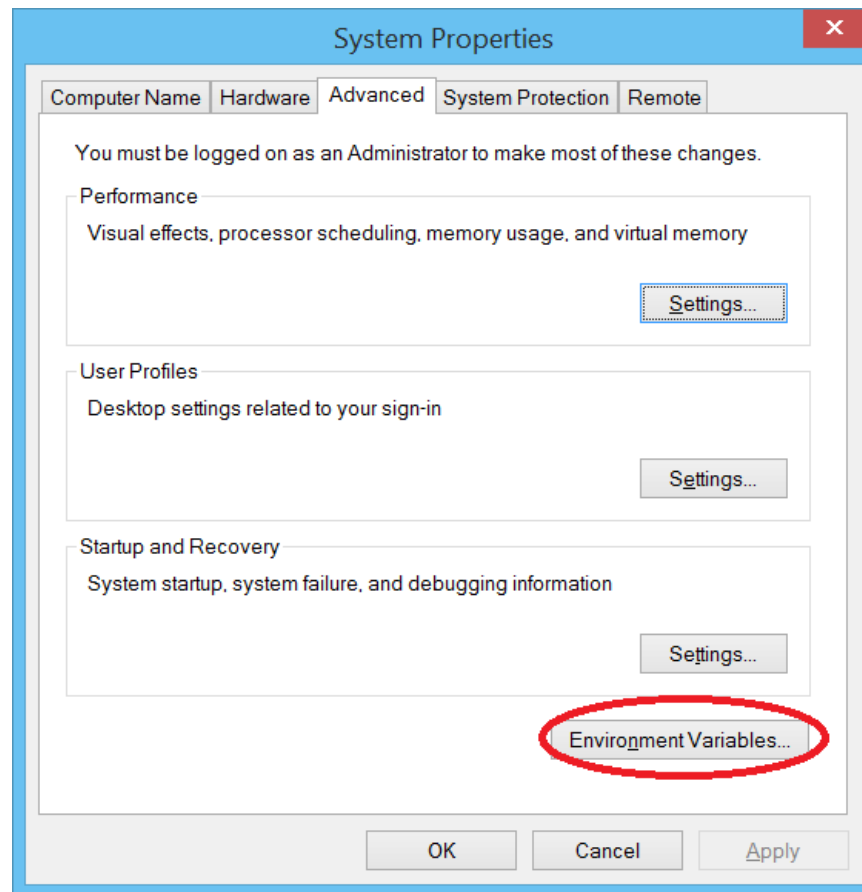


- b) On the left-hand side of the System window click on 'Advanced System Settings':





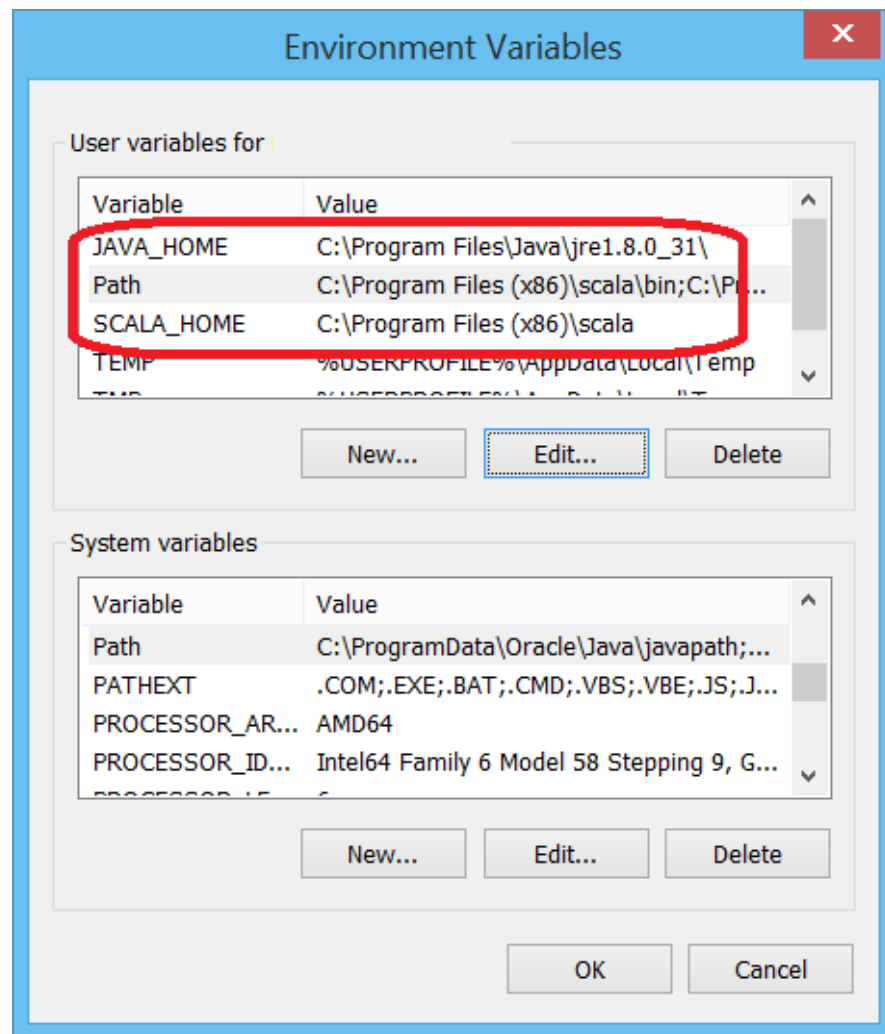
- c) Click on the 'Environment Variables...' button of the Advanced tab in the System Properties window:





- d) Review the User Variables list and make the appropriate changes:
- Check, whether the **JAVA_HOME** variable is defined and points to the correct version of Java.
 - Create or check the **SCALA_HOME** variable pointing to your Scala directory.
 - Prepend your user **Path** variable with the **%SCALA_HOME%\bin;** (remember the trailing semicolon).

After the changes, your Environment Variables dialog should look like this:



- e) The **%SCALA_HOME%** variable may or may not be expanded in the **Path** variable value, depending on the current system state. In either case it will be interpreted correctly. Thus, your **Path** user variable can start either **%SCALA_HOME%\bin;**, or something like: **C:\Program Files (x86)\scala\bin** in the User Variables table.
- Close Environment Variables and System Properties windows.



- ii. Close all command line windows. This is important for your Environment Variables changes to take effect!
4. Check that the newly installed Scala is working properly:
- a) Open a new Command Line window (E.g. press <Win-R> and in the Search window type **cmd**).
 - b) Type **scala** and press <ENTER>.
 - c) Scala should run and **scala>** prompt should appear at the last line:

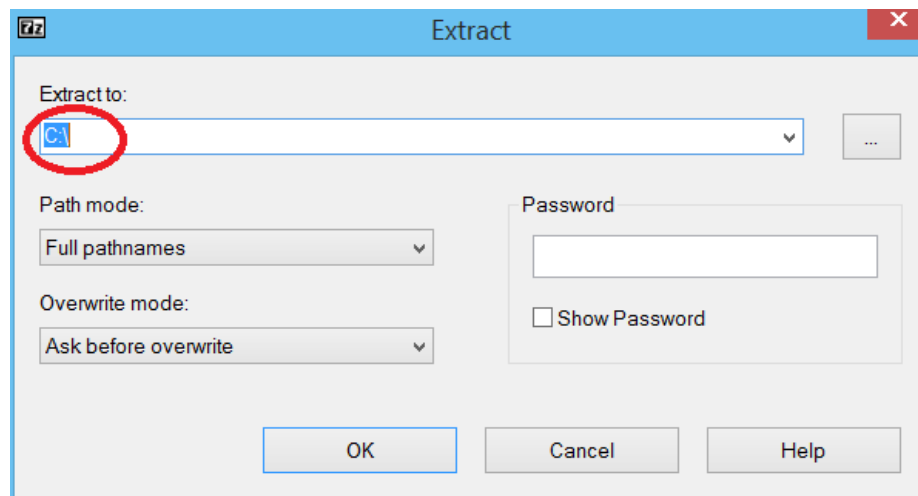
```
C:\>scala
Welcome to Scala version 2.10.5 (Java HotSpot(TM) 64-Bit Server VM, Java 1.8.0_31).
Type in expressions to have them evaluated.
Type :help for more information.
scala>
```

- d) Type **:q** and press <ENTER>. Scala will quit.
- Congratulations! Your Scala installation is up and functional.

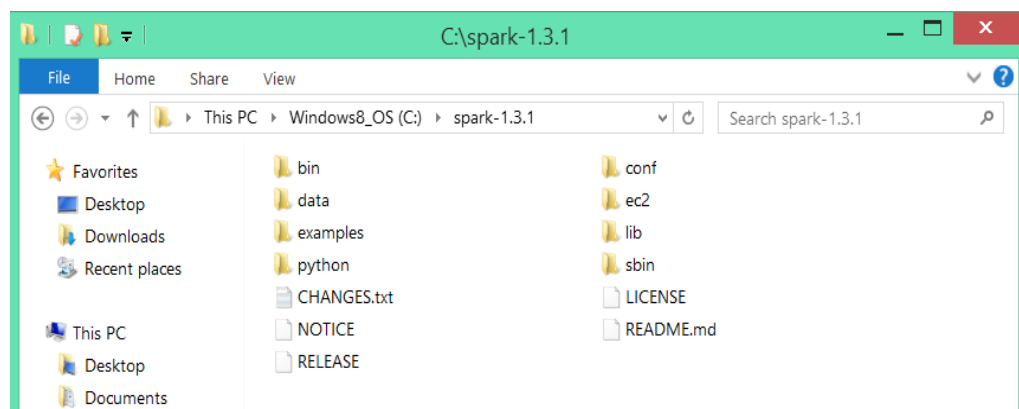


Install Spark

1. Unpack the downloaded Spark installation archive. You can unpack Spark into any location in your computer. The guide below assumes you have unpacked Spark into the root folder of the C: drive of your computer. Please update all paths, settings and variables provided in this guide to reflect your location. The guide describes how to use the 7-zip archiver mentioned earlier to unpack the archive:
 - a) Right-click the downloaded Spark package and select '7-Zip → Extract Here'. A new file **spark-x.y.z-bin-hadoop2.6** is created in the download directory, **x.y.z** represents your Spark version.
 - b) Right-click the newly created file and select '7-zip → Extract files...'. In the Extract dialog change the content of the 'Extract to:' field to **C:**
 - c)



- d) Press OK. The files are extracted. When extraction is complete, rename the folder **C:\spark-x.y.z-bin-hadoop2.6** to **C:\spark-x.y.z**, replace **x.y.z** with your version of Spark.
- e) Check the contents of the **C:\spark-x.y.z** folder. It should look like this:





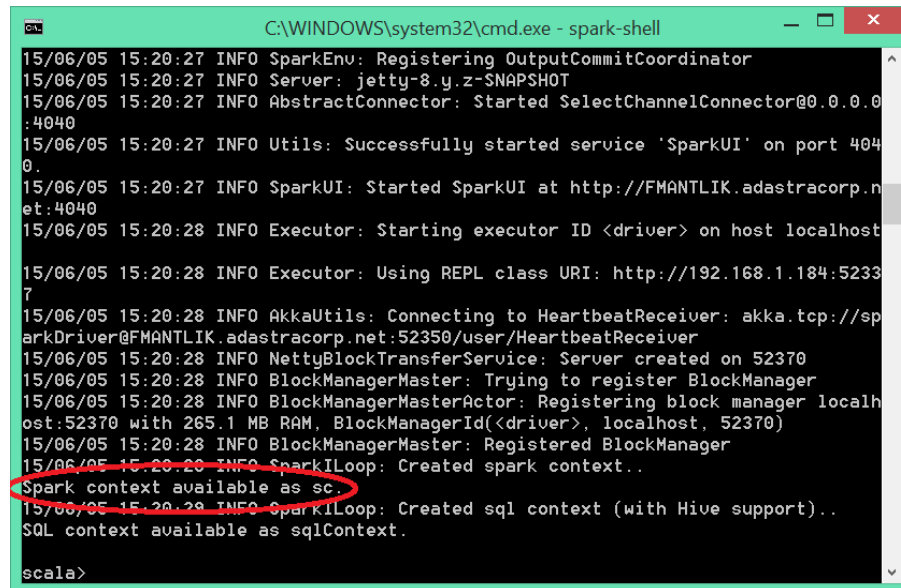
2. Although we are not using Hadoop with Spark, when running Spark an error will be reported if the **HADOOP_HOME** variable is not configured. To overcome this bug we need to create a fake Hadoop installation directory and populate it with the **winutils.exe** file downloaded in the Download section on page 4:
 - a) Create folders **C:\winutils** and **C:\winutils\bin**
 - b) Copy the downloaded file **winutils.exe** to **C:\winutils\bin**
3. Setup required environment variables:
 - a) Using a procedure similar to the one we used when setting up Install Scala, set up the following user environment variables:
 - i. **SPARK_HOME** - point to the Spark installation directory, e.g. **C:\spark-1.3.1**
 - ii. **Path** – prepend the user variable **Path** with **%SPARK_HOME%\bin;**
 - iii. **HADOOP_HOME** – **C:\winutils**.
 - b) The following user environment variables are needed for Scala and Spark to work properly:

Variable	Value
HADOOP_HOME	c:\winutils
JAVA_HOME	<your-Java-installation-directory>
SCALA_HOME	<your-Scala-installation-directory>
SPARK_HOME	c:\spark-x.y.z
Path	%SCALA_HOME%/bin;%SPARK_HOME%/bin;...



4. Check your new Spark installation:

- a) Open command window
- b) Type **spark-shell** and press <ENTER>. Wait until Spark is up and running. It should take less than a minute. The command window displays lots of logging messages. Wait for the Scala prompt **scala>** and note the highlighted sentence:



```
C:\WINDOWS\system32\cmd.exe - spark-shell
15/06/05 15:20:27 INFO SparkEnv: Registering OutputCommitCoordinator
15/06/05 15:20:27 INFO Server: jetty-8.y.z-SNAPSHOT
15/06/05 15:20:27 INFO AbstractConnector: Started SelectChannelConnector@0.0.0.0:4040
15/06/05 15:20:27 INFO Utils: Successfully started service 'SparkUI' on port 4040.
15/06/05 15:20:27 INFO SparkUI: Started SparkUI at http://FMANTLIK.adastracorp.net:4040
15/06/05 15:20:28 INFO Executor: Starting executor ID <driver> on host localhost
15/06/05 15:20:28 INFO Executor: Using REPL class URI: http://192.168.1.184:52337
15/06/05 15:20:28 INFO AkkaUtils: Connecting to HeartbeatReceiver: akka.tcp://sparkDriver@FMANTLIK.adastracorp.net:52350/user/HeartbeatReceiver
15/06/05 15:20:28 INFO NettyBlockTransferService: Server created on 52370
15/06/05 15:20:28 INFO BlockManagerMaster: Trying to register BlockManager
15/06/05 15:20:28 INFO BlockManagerMasterActor: Registering block manager localhost:52370 with 265.1 MB RAM, BlockManagerId(<driver>, localhost, 52370)
15/06/05 15:20:28 INFO BlockManagerMaster: Registered BlockManager
15/06/05 15:20:28 INFO SparkILoop: Created spark context..
Spark context available as sc.
15/06/05 15:20:29 INFO SparkILoop: Created sql context (with Hive support)..
SQL context available as sqlContext.

scala>
```

- c) Open an internet browser and connect to the Spark server at the address <http://localhost:4040> . There is not much to see here because no Spark jobs are currently running. You can inspect the configuration parameters under the 'Environment' tab and watch the Spark executor ready to run jobs under the 'Executors' tab.
- d) Type **:q** in the command window to exit Scala and quit Spark. The Spark server is stopped and no longer available.

5. Run built-in Spark test:

- a) At the command window type **run-example SparkPi 10** and press <ENTER>. **NOTE:** The command is case sensitive.
- b) The command starts Spark and runs a job which computes an approximate value of Pi. You can watch the computation progress at <http://localhost:4040> .
- c) When the computation is finished, scroll up in the output window and locate the line with the result:



```
C:\WINDOWS\system32\cmd.exe

109 ms on localhost (7/10)
15/06/05 15:58:04 INFO Executor: Finished task 6.0 in stage 0.0 (TID 6). 736 bytes result sent to driver
15/06/05 15:58:04 INFO TaskSetManager: Finished task 6.0 in stage 0.0 (TID 6) in 141 ms on localhost (8/10)
15/06/05 15:58:04 INFO Executor: Finished task 9.0 in stage 0.0 (TID 9). 736 bytes result sent to driver
15/06/05 15:58:04 INFO Executor: Finished task 7.0 in stage 0.0 (TID 7). 736 bytes result sent to driver
15/06/05 15:58:04 INFO TaskSetManager: Finished task 9.0 in stage 0.0 (TID 9) in 110 ms on localhost (9/10)
15/06/05 15:58:04 INFO TaskSetManager: Finished task 7.0 in stage 0.0 (TID 7) in 125 ms on localhost (10/10)
15/06/05 15:58:04 INFO DAGScheduler: Stage 0 (reduce at SparkPi.scala:35) finished in 26.391 s
15/06/05 15:58:04 INFO TaskSchedulerImpl: Removed TaskSet 0.0, whose tasks have all completed, from pool
15/06/05 15:58:04 INFO DAGScheduler: Job 0 finished: reduce at SparkPi.scala:35, took 27.277204 s
Pi is roughly 3.143852
15/06/05 15:58:04 INFO ContextHandler: stopped o.s.j.s.ServletContextHandler(/metrics/json,null)
15/06/05 15:58:04 INFO ContextHandler: stopped o.s.j.s.ServletContextHandler(/stages/stage/kill,null)
15/06/05 15:58:04 INFO ContextHandler: stopped o.s.j.s.ServletContextHandler(/,null)
```

NOTE: your assessed value of Pi will likely differ from the example.



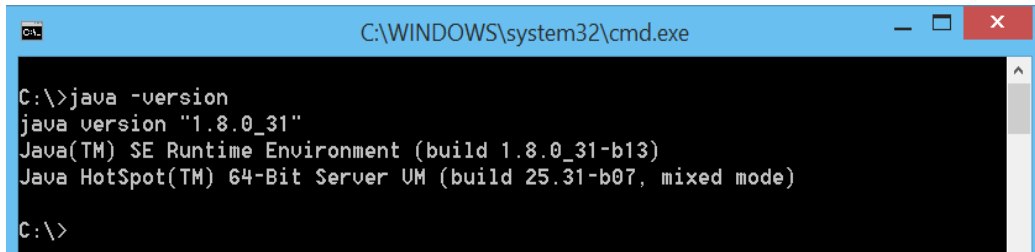
Prepare training data

1. Unpack the archive with training data into the folder ***c:\sparktraining***. You can use another folder but you will need to adjust data path accordingly when running the Spark examples in this course.



Appendix: Java installation troubleshooting

Open Command window and type **java -version** followed by <ENTER>. The output should look like this:



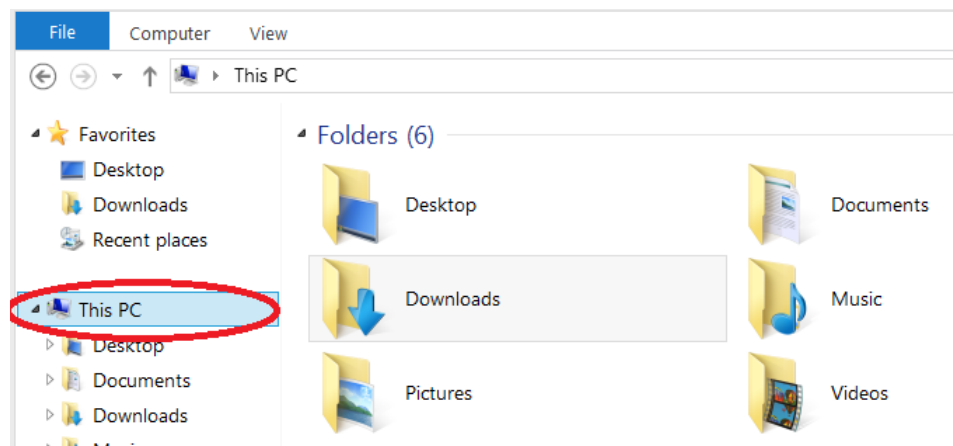
```
C:\WINDOWS\system32\cmd.exe

C:\>java -version
java version "1.8.0_31"
Java(TM) SE Runtime Environment (build 1.8.0_31-b13)
Java HotSpot(TM) 64-Bit Server VM (build 25.31-b07, mixed mode)

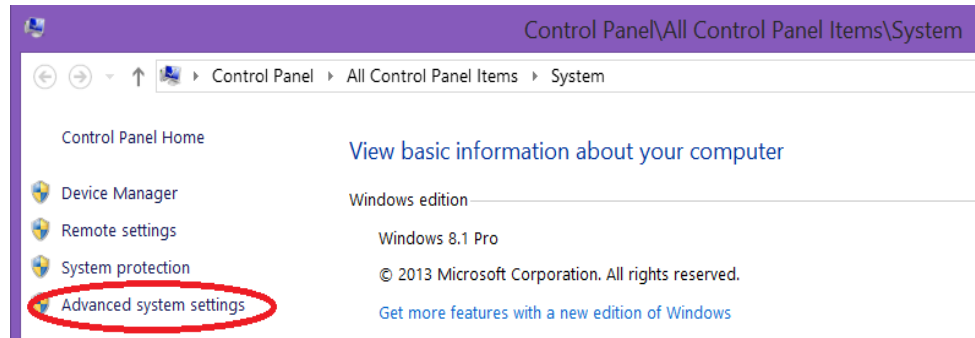
C:\>
```

If your output is different, there is a problem with the Java installation. Follow these steps to fix the problem:

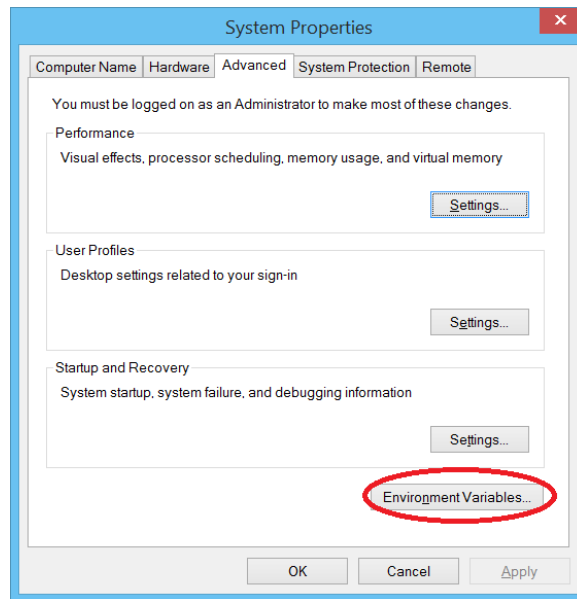
1. Uninstall all old versions of Java.
2. Visit <http://java.com/download>.
3. Click on 'All Java downloads' link from the left menu.
4. Download proper Offline installer for your version of Windows. You must use 'Windows Offline' installer on 32-bit version and 'Windows Offline (64-bit)' on 64-bit version.
5. Run the downloaded installer and follow the on-screen instructions.
6. Locate your Java installation. With File Explorer navigate to **c:\Program Files\Java** and note the folder where Java was installed. It will be similar to **jre1.8.0_31**.
7. Change the user environment variables **JAVA_HOME** and **Path** following these steps:
 - a. In File Explorer, right-click 'This PC' icon and select 'Properties':



- b. On the left-hand side of the System window click on 'Advanced System Settings':



- c. Click on the 'Environment Variables...' button of the 'Advanced' tab in the 'System Properties' window:



- d. Review the 'User Variables' list and make the appropriate changes. The **JAVA_HOME** user variable must point to your Java installation directory, e.g. **C:\Program Files\Java\jre1.8.0_31**. Prepend your **Path** variable with the following: **%JAVA_HOME%\bin;** Do not forget the semicolon.
8. Close 'Environment Variables' and 'System Properties' windows.
 9. Close all command line windows. This is important for your environment variables changes to take effect!
 10. Re-open command line window and check the Java installation with the command **java -version**.